

P1f: zeros of B at every root of unity of the middle arc,
density of the poles of G_0^T , and the transfer to U and V
research programme P1, sixth atom, 2026-09-24. UNREVIEWED.

Notation of P1d.lemma.tex and P1e.lemma.tex; P1d numbering as compiled (Theorem 2 Gauss factorisation, Theorem 5 Fresnel, Theorem 9 corrected Lemma (1''- Δ)). Paper templates: paper2a.tex, Propositions `prop:minusone`, `prop:t1minusone`, `prop:qi`, Corollaries `cor:Uminusone`, `cor:Ui`. Labels: PROVED (complete proof here, at the level of rigour of `prop:minusone`: the $O(t)$ of Laplace's method and the index j_0 are not made explicit), *computer-assisted* (reduced to finitely many Arb inequalities checked by the named script, which fails loudly), VERIFIED (numerics, not a proof), HEURISTIC, OPEN.

Outcome.

- (A) The four editorial gaps of Reviewer AA in P1e.lemma.tex are fixed (§1).
- (B) **Theorem 9 (PROVED; computer-assisted only through Theorem M of P1e).** Let $\zeta = e(a/N)$, $\gcd(a, N) = 1$, $\frac{1}{5} \leq \frac{a}{N} \leq \frac{4}{5}$, $N \geq 16$. Then the zeros of $B(q) = 1 - \Sigma_1(q) = \cos(Z; q^2)$ accumulate at $q = \zeta$ along an explicit tie ray, they are simple, and they are simple poles of G_0^T . **Corollary 10 (PROVED):** the poles of G_0^T accumulate at every point of the arc $\arg q \in [\frac{2\pi}{5}, \frac{8\pi}{5}]$; in $x = q^{1/2}$, at every point of the arcs $\arg x \in [\frac{\pi}{5}, \frac{4\pi}{5}]$ and $\arg x \in [\frac{6\pi}{5}, \frac{9\pi}{5}]$. No transfer-model standing is needed for G_0^T .
- (C) **Transfer to U , V : reduced to two explicit finite conditions, NOT closed for a dense set.** Theorem 11 (PROVED): at the zeros of Theorem 9, $S_e \rightarrow$ a multiple of $Z_N^+ := \sum_k \psi_k \zeta^{-k^2-k}$ and $t_1 \rightarrow t_1^* = -\frac{1}{2}(1 + \omega)$ with $\omega = Z_N^-/(u_0 Z_N^+)$, for every $N \geq 16$ in the family. So U, V have poles accumulating at $\pm\sqrt{\zeta}, \pm\sqrt{\bar{\zeta}}$ as soon as $Z_N^+ \neq 0$ and ω avoids six explicit values (standing: Theorems `thm:model`, `thm:RJ`). Both conditions are certified in Arb for all 4116 roots of unity with $3 \leq N \leq 150$ in the family except $\zeta = -1$ (`P1f_transfer_cert.py`); combined with Theorem 9, the poles of U and V are PROVED (computer-assisted) to accumulate at $\pm\sqrt{\zeta}, \pm\sqrt{\bar{\zeta}}$ for every such ζ with $16 \leq N \leq 150$. **OPEN:** $Z_N^+ \neq 0$ and the ω -conditions for infinitely many N forming a dense set; hence *density* of the poles of U and V on the arcs is OPEN. The formula $t_1^* = -\frac{1}{2}(1 + \omega)$ reproduces the published $q = i$ value $t_1^* = 0.07275 - 0.57275i$ of `cor:Ui` exactly (VERIFIED).

Remark on the brief: $a/N \in [\frac{1}{5}, \frac{4}{5}]$ is $\arg q \in [\frac{2\pi}{5}, \frac{8\pi}{5}]$, which is $\arg x \in [\frac{\pi}{5}, \frac{4\pi}{5}]$ (and its negative x -image), not $[\frac{\pi}{10}, \frac{2\pi}{5}]$ as written in the brief.

1 Task A: the Reviewer AA gaps in P1e.lemma.tex

- *Lemma H, case $m' = n$* (PROVED, written into P1e.lemma.tex). In (2) the reduced fraction i'/m' may have $m' \leq n$. For $m' < n$ the old shallower-hazard sentence applies. For $m' = n$, $i' \neq j$ gives $|x - i'/n| \geq 1/(2n)$; $i' = j$ gives $D = |d| < y^m/(4\tau)$ with $m \geq 1/(2nD)$, i.e.

$D 2^{1/(2nD)} < 12.5$, which contradicts $f(D) = D 2^{1/(2nD)} \geq f(\varepsilon_n) \geq 12.5$ (f decreasing on $(0, \log 2/(2n)]$; $2^n \geq 25n^2$ for $n \geq 53$, $n 2^n/4 \geq n + 2 \log_2 n + 3$ for $10 \leq n \leq 52$).

- *Directed rounding* (`dround.py`; `P1e_cert.py` and `P1e_lowseed.py` now print lower bounds rounded down and upper bounds rounded up; c_{seed} is recomputed in Arb from the rounded-down $|S_{\text{amp}}|$). All 18 low seeds re-certified; the low-seed table of P1e replaced. Smallest $c_{\text{seed}} = 0.1921$ (seed 3/4), largest $\kappa_{\text{seed}} = 0.00132$; the conclusions of Theorem M are unchanged.
- *Fail-loud base scan*: `P1e_logderiv.py scan 2 150 arc` selects $N \leq 5a \leq 4N$ by an integer test and aborts with exit status 1 and `SCAN FAILED` on any exception or non-certified $|Z| > 0$ (tested by injecting a failure at 2/5). Rerun: 4117 cases, all pass, $\min |Z_N| \geq 0.9134$.
- *Cross-references*: P1d Theorems 1, 3, 5 of the old text are Theorems 2, 5, 9 of the compiled P1d; Lemmas W, conj, dec, lap are Lemmas 1, 4, 6, 8. Corrected throughout P1e_lemma.tex.

2 Setting and the saddle data

Fix $\zeta = e(a/N)$ in the family, $N \geq 3$, $\zeta \neq -1$. Put $c = -2(1 - \zeta)$, $L = \text{Log } c = \ell + i\varphi$ ($\ell \geq \ell_0 := \log(4 \sin \frac{\pi}{5}) = 0.85483$, $|\varphi| \leq \pi$), w the root of $c^N w = (1 - w)^2$ with $|w| \leq 2|c|^{-N}$ (Lemma 1 of P1d), $\xi = \frac{1}{N} \text{Log}(1 - w)$, $u_0 = e^{2\xi}/c = \lambda^2/c$, $\text{lin} = \zeta/(1 - \zeta)$, and $q = \zeta e^{-t}$, $t = re^{i\theta}$, $p = e^{-t}$. Write $b_k = c_t^k q^{k^2}/(q; q)_{2k}$, $c_t = -2(1 - q)$, so $B = \sum_k b_k$. For $m \in \mathbb{Z}$ let

$$L_m = L + \frac{2\pi i m}{N}, \quad \psi_m = \varphi + \frac{2\pi m}{N}, \quad x_m = \frac{L_m}{2} - \xi, \quad \Phi(x) = xL - x^2 - \frac{\text{Li}_2(e^{-2Nx})}{N^2} + \frac{\pi^2}{6N^2}, \quad \Omega_m(x) = \Phi(x) + \frac{2\pi i m x}{N},$$

$$V_m = \Omega_m(x_m), \quad h_m(\theta) = \text{Re}(V_m e^{-i\theta}), \quad G^*(m) = \sum_{r \bmod N} e((ar^2 - mr)/N).$$

Lemma 1 (saddles; PROVED). (i) $\Omega'_m(x_m) = 0$ and $\Omega''_m(x_m) = -2\frac{1+w}{1-w}$. (ii) $V_m = \frac{L_m^2}{4} + E_w$ with $E_w = \frac{\pi^2}{6N^2} - \xi^2 - \frac{\text{Li}_2(w)}{N^2}$ independent of m ; hence $h_m(\theta) = \frac{1}{4} \left[\frac{\ell^2}{\cos \theta} - \cos \theta (\psi_m - \ell \tan \theta)^2 \right] + \text{Re}(E_w e^{-i\theta})$, a concave quadratic in ψ_m . (iii) $\Omega_m(x) = V_m - (x - x_m)^2 - R_m(x)$ with $R_m(x) = N^{-2} [\Lambda(2Nx) - \Lambda(2Nx_m) - \Lambda'(2Nx_m) 2N(x - x_m)]$, $\Lambda(v) = \text{Li}_2(e^{-v})$, $\Lambda'(v) = \text{Log}(1 - e^{-v})$, $\Lambda''(v) = 1/(e^v - 1)$. For $\text{Re } x > 0$: $|R_m(x)| \leq 2|x - x_m|^2 \sup_{y \in [x, x_m]} |e^{2Ny} - 1|^{-1}$ and $|R_m(x)| \leq N^{-2} (\text{Li}_2(e^{-2N \text{Re } x}) + 1.1|w|(1 + 2N|x - x_m|))$. (iv) $G^*(m) \neq 0$ iff $m \in \mathcal{M}$, where $\mathcal{M} = \mathbb{Z}$ (N odd), $2\mathbb{Z}$ ($4 \mid N$), $2\mathbb{Z} + 1$ ($N \equiv 2 \pmod{4}$); then $|G^*(m)| = \sqrt{N}$ resp. $\sqrt{2N}$. For N odd, $G^*(m) = e(-4am^2/N) G^*(0)$ ($\ell = \text{inverse mod } N$).

Proof. (i) $\Omega'_m(x) = L_m - 2x - \frac{2}{N} \text{Log}(1 - e^{-2Nx})$ and $e^{-2Nx_m} = c^{-N}(1 - w)^2 = w$. (ii) Expand $x_m L_m - x_m^2$. (iii) Taylor at $v = 2Nx_m$, using $\Lambda'(2Nx_m) = \text{Log}(1 - w) = N\xi$; the second bound is the triangle inequality with $|\text{Li}_2(z)| \leq \text{Li}_2(|z|)$, $|\text{Log}(1 - w)| \leq 1.1|w|$. (iv) Classical quadratic Gauss sums (as in Theorem 2 of P1d). $\square$

VERIFIED: the V_m , the family x_m and $|A(n)/A(n+1)| = e^{\pi \text{Im}(\text{lin})/N}$ agree with the scoping script `k1.py`.

The tie ray. Let $s = 1$ (N odd), $s = 2$ (N even), and choose $n \in \mathcal{M}$ minimising $|\psi_n + \pi s/N|$; so $n, n + s$ are consecutive in $\mathcal{M}$. Define θ^* by $\tan \theta^* = (\psi_n + \pi s/N)/\ell$. Then $|\tan \theta^*| \leq \pi s/(N\ell)$, $h_n(\theta^*) = h_{n+s}(\theta^*) =: T$,

$$T = \frac{\ell^2}{4 \cos \theta^*} - \frac{\pi^2 s^2 \cos \theta^*}{4N^2} + \text{Re}(E_w e^{-i\theta^*}), \quad h_m(\theta^*) \leq T - \frac{2\pi^2 s^2 \cos \theta^*}{N^2} \quad (m \in \mathcal{M} \setminus \{n, n + s\}).$$

Numerical facts for $N \geq 16$ (PROVED by the displayed elementary estimates): $|\tan \theta^*| \leq 0.4594$, $|\theta^*| \leq 24.7^\circ$, $\cos \theta^* \geq 0.9087$; $|w| \leq 2e^{-16\ell_0} \leq 2.3 \cdot 10^{-6}$; $T \geq \frac{\ell_0^2}{4} - \frac{\pi^2(s^2/4 - \cos \theta^*/6)}{N^2} - 10^{-8} \geq 0.1499$.

3 The amplitude

Lemma 2 (PROVED). *Let $d_{r,j} = (2r - j) \bmod N$ and, for $m \in \mathbb{Z}$,*

$$A(m) = e^{x_m \text{lin}} \sum_{r \bmod N} \zeta^{r^2} e(-\frac{mr}{N}) \prod_{j=1}^N (1 - \zeta^j e^{-2x_m})^{-(\frac{1}{2} - \frac{d_{r,j}}{N})} \quad (\text{principal powers}).$$

Then $A(m) = e^{x_m \text{lin}} (1 - w)^{-1/(2N)} G^(m) Z_N$, with $Z_N = \sum_k \psi_k \zeta^{-k^2}$ the object of Theorem 2 of P1d and Theorem M of P1e. In particular $A(m) \neq 0$ for every $m \in \mathcal{M}$, and $A(m) = 0$ for $m \notin \mathcal{M}$. More generally, with $\zeta^{\pm r}$ inserted in the r -sum and $e^{x_m \text{lin}}$ replaced by $e^{x_m(\text{lin} \mp 1)}$ (the amplitude $A^\pm(m)$ of $S_\pm$ below), $A^\pm(m) = e^{x_m(\text{lin} \mp 1)} (1 - w)^{-1/(2N)} G^*(m \mp a) Z_N^\pm$, $Z_N^\pm := \sum_k \psi_k \zeta^{-k^2 \mp k}$.*

Proof. $e^{-2x_m} = u_0 e(-m/N)$ with $|u_0| < 1$. By Lemma tel of S0 lemma, applied termwise in r (its proof is termwise), the product equals $(1 - w)^{-1/(2N)} F(\zeta^{2r} u_0 e(-m/N))$ with $F(v) = \sum_k h_k v^k$, $h_k u_0^k = \psi_k$, absolutely convergent on $|v| = |u_0|$. Then $\sum_r e((ar^2 + 2akr - (m \mp a')r)/N)$, $a' \in \{0, \pm a\}$, becomes $e((-ak^2 + (m \mp a')k)/N) G^*(m \mp a')$ after $r \mapsto r - k$; the factor $e(-mk/N)$ from e^{-2kx_m} cancels the m -dependence, leaving $\zeta^{-k^2 \mp k}$. Nonvanishing: Lemma 1(iv) and Theorem M ($Z_N \neq 0$ on the whole family). $\square$

VERIFIED: P1f_model.py: $B(\zeta e^{-t})$ divided by $\kappa \sum_m A(m) e^{V_m/t}$ (Theorem 8) is $0.9992 - 0.0088i$, $1.0041 - 0.0029i$ at $1/3$ ($r = 0.05, 0.02$), $1.0121 - 0.0098i$, $1.0042 - 0.0035i$ at $2/5$, $1.0076 + 0.0040i$, $1.0039 + 0.0020i$ at $3/10$ ($r = 0.03, 0.015$): the error is $O(t)$.

4 Exact decomposition

Let $N' = N$ (N odd) and $N' = N/2$ (N even), and $\epsilon = -1$ if $N \equiv 2 \bmod 4$, else $\epsilon = 1$. For $0 \leq \rho < N'$ and $k \in \mathbb{C}$ put

$$F_\rho(k) = \zeta^{\rho^2} \exp(k \text{Log } c_t - tk^2) \prod_{\iota=1}^N (\zeta^{2\rho+\iota} e^{-t(2k+\iota)}; p^N)_\infty / (q; q)_\infty.$$

Lemma 3 (PROVED). *F_ρ is entire and $b_{N'j+\rho} = \epsilon^j F_\rho(N'j + \rho)$ for $j \geq 0$.*

Proof. $(q; q)_{2k} = (q; q)_\infty / (q^{2k+1}; q)_\infty$ and $(q^{2k+1}; q)_\infty = \prod_{\iota=1}^N (\zeta^{2k+\iota} p^{2k+\iota}; p^N)_\infty$; $\zeta^{2(N'j+\rho)} = \zeta^{2\rho}$ since $N \mid 2N'$. Finally $\zeta^{(N'j+\rho)^2} = \zeta^{\rho^2} \zeta^{2N'j\rho} e(aN'^2 j^2 / N)$ with $\zeta^{2N'j\rho} = 1$. If N is odd, $N'^2/N = N$ and the last factor is 1. If N is even, it is $e(aN'j^2/2)$: 1 when N' is even ($4 \mid N$), and $(-1)^{j^2} = (-1)^j$ when N' and a are odd ($N \equiv 2 \bmod 4$). $\square$

Fix $\delta = \frac{1}{8N}$, $K = \lfloor \delta/r \rfloor$, $H = \sum_{k \leq K} b_k$, and for each ρ let j_ρ be the least j with $N'j + \rho > K$, $s_\rho = j_\rho - \frac{1}{2}$, $g_\rho(s) = F_\rho(N's + \rho)$. In the coordinate $x = kt$ the point s_ρ is $x_A^\rho = (N's_\rho + \rho)t$, which lies on the ray $\arg x = \theta$ with $|x_A^\rho| \in [\delta/2, 2\delta]$ for $r \leq \delta/(2N)$. With $\gamma_\mp^\rho$ the rays $x_A^\rho + e^{i(\theta \mp \pi/8)}[0, \infty)$ (in the s -plane they leave the half-integer s_ρ into $\mp \text{Im } s > 0$ at angle $\pi/8$), the residue theorem gives exactly (as Step 1 of prop:minusone for $\epsilon = 1$ and Step 1 of prop:tminusone for $\epsilon = -1$)

$$B = H + \sum_{\rho < N'} \left(\int_{\gamma_-^\rho} g_\rho \mathcal{K}_- ds + \int_{\gamma_+^\rho} g_\rho \mathcal{K}_+ ds \right),$$

where for $\epsilon = 1$ $\mathcal{K}_- = (1 - e^{-2\pi i s})^{-1}$, $\mathcal{K}_+ = e^{2\pi i s} (1 - e^{2\pi i s})^{-1}$; for $\epsilon = -1$, $\mathcal{K}_\mp = e^{\mp i \pi s} (1 - e^{\mp 2\pi i s})^{-1}$. Decay at the ends and on the closing arcs holds because $|\theta| < \pi/4$ (so all directions between $\theta - \pi/8$

and $\theta + \pi/8$ lie in the Gaussian sector $|\arg x - \theta/2| < \pi/4$) and the product is bounded on $\operatorname{Re} x > 0$. Expanding the kernels $M = 2N$ times,

$$\mathcal{K}_- = \sum_{\nu \in I_-} e^{2\pi i \nu s} + e^{2\pi i \nu_- s} \frac{1}{1 - e^{-2\pi i s}}, \quad \mathcal{K}_+ = \sum_{\nu \in I_+} e^{2\pi i \nu s} + e^{2\pi i \nu_+ s} \frac{1}{1 - e^{2\pi i s}},$$

with $I_- = \{0, -1, \dots, -(M-1)\}$, $\nu_- = -M$, $I_+ = \{1, \dots, M\}$, $\nu_+ = M+1$ for $\epsilon = 1$, and, for $\epsilon = -1$, all of these shifted by $-\frac{1}{2}$: $I_- = \{-\frac{1}{2}, \dots, -(M-\frac{1}{2})\}$, $\nu_- = -M-\frac{1}{2}$, $I_+ = \{\frac{1}{2}, \dots, M-\frac{1}{2}\}$, $\nu_+ = M+\frac{1}{2}$ (from $\mathcal{K}_\pm = \sum_{k \geq 0} e^{\pm 2\pi i (k+\frac{1}{2})s}$; corrected 2026-09-24, the earlier text shifted I_+ by $+\frac{1}{2}$ and so omitted $\nu = \frac{1}{2}$). Hence every explicit mode has $|\mu| \leq 2N$ (N odd) resp. $|\mu| \leq 4N+1$ (N even), and the remainder modes are $\mu_\mp = \mp 2N, 2N+1$ (N odd), $\mp 4N, 4N+2$ ($\epsilon = 1, N$ even), $\mp(4N+1)$ ($\epsilon = -1$). The mode ν has x -frequency $\mu = \nu N/N' \in \mathcal{M}$: $e^{2\pi i \nu s} = e(-\mu\rho/N) e^{2\pi i \mu x/(Nt)}$. Each explicit-mode integral has an entire integrand and is moved, with fixed endpoints, to

$$\Pi_\mu^\rho = [x_A^\rho, x_\mu] \cup (x_\mu + e^{i\theta/2}[0, \infty)).$$

5 Euler–Maclaurin, head, landscape

Lemma 4 (EM; PROVED). *Fix N , $\delta' > 0$, $\theta_{\max} < \pi/2$. Uniformly for $|\theta| \leq \theta_{\max}$, $0 < r \leq r_0$, $\operatorname{Re} x \geq \delta'$ and every ρ ,*

$$\log F_\rho(x/t) = \frac{\Phi(x)}{t} + a_\rho(x) + \frac{1}{2} \log \frac{Nt}{2\pi} + O(|t|(1+|x|)),$$

$$a_\rho(x) = x \ln + \frac{2\pi i a \rho^2}{N} - \sum_{j=1}^N \left(\frac{1}{2} - \frac{d_{\rho j}}{N}\right) \operatorname{Log}(1 - \zeta^j e^{-2x}) - \sum_{i=1}^{N-1} \left(\frac{1}{2} - \frac{i}{N}\right) \operatorname{Log}(1 - \zeta^i).$$

Proof. As Step 2 of `prop:minusone` and the ray lemma of `lem:qi-bounds`. For each ι , $\sum_{M \geq 0} G(2x + \iota t + NtM)$ with $G(v) = \operatorname{Log}(1 - \zeta^{2\rho+\iota} e^{-v})$ equals $-\frac{\operatorname{Li}_2(\zeta^{2\rho+\iota} e^{-2x-\iota t})}{Nt} + \frac{1}{2}G(2x + \iota t) + E_0$, $|E_0| \leq \frac{N|t|}{3}(\frac{1}{\delta_d} + \frac{1}{\delta_d^2 \cos \theta})$, where along the ray $|1 - \zeta^{2\rho+\iota} e^{-v}| \geq \delta_d := 1 - e^{-2\delta'}$. Taylor in ιt ($\frac{d}{dv} \operatorname{Li}_2(\zeta e^{-v}) = \operatorname{Log}(1 - \zeta e^{-v})$) gives $-\frac{\operatorname{Li}_2(\zeta e^{-2x})}{Nt} + (\frac{1}{2} - \frac{\iota}{N}) \operatorname{Log}(1 - \zeta^{2\rho+\iota} e^{-2x}) + O(t)$; with $j \equiv 2\rho + \iota$, $\frac{1}{2} - \frac{\iota}{N} = -(\frac{1}{2} - \frac{d_{\rho j}}{N})$, and $\sum_j \operatorname{Li}_2(\zeta^j z) = \frac{1}{N} \operatorname{Li}_2(\zeta^N z)$. $(q; q)_\infty = \prod_{i=1}^N (\zeta^i p^i; p^N)_\infty$: for $i < N$ the ray lemma from $v = \iota t$ (distance $\geq \mathcal{D}(0, \operatorname{dist}(2\pi i a/N, 2\pi \mathbb{Z})) > 0$), for $i = N$ the η -transformation; $\sum_{i < N} \operatorname{Li}_2(\zeta^i) = \frac{\pi^2}{6}(\frac{1}{N} - 1)$. Finally $\operatorname{Log} c_t = L + t \ln + O(t^2)$. $\square$

Lemma 5 (head; PROVED). *For $N \geq 16$ and $0 \leq k \leq K$: $r \log |b_k| \leq H_{\text{head}} + r \log(1/P_N)$ with $H_{\text{head}} = \delta(\ell + 0.0837 + r) + \frac{2\delta}{N}(1 + \log \frac{1}{1.8\delta}) \leq 0.0167 + r$, where $P_N := \min_{0 \leq J < N} \prod_{i \leq J} |1 - \zeta^i| > 0$ is the least partial product over one period (no factor 0.959 in it; see the proof).*

Proof. $|q^{k^2}| \leq 1$; $|c_t| \leq |c| + 2r(1.01)$, so $\log |c_t| \leq \ell + r$. For $i \leq 2k \leq 2\delta/r$, $N \nmid i$: $|1 - e^{-it}| \leq \frac{1}{4N} e^{1/64} \leq \frac{0.254}{N}$ and $|1 - \zeta^i| \geq 2 \sin \frac{\pi}{N} \geq \frac{6.24}{N}$, so $|1 - \zeta^i e^{-it}| \geq 0.959|1 - \zeta^i|$. *Bookkeeping of the 0.959*: there are at most $2k \leq 2\delta/r$ such indices, so these factors cost at most $0.959^{2\delta/r}$, i.e. $2\delta(-\log 0.959)$ in $r \log |b_k|$. Precisely: the per-factor ratio is $\geq 1 - 0.254/6.24 = 0.95929$, and $2(-\log 0.95929) = 0.08312 \leq 0.0837$ (with the rounded 0.959 one would get 0.08373, which exceeds 0.0837 by $3 \cdot 10^{-5}$; the unrounded ratio is what is used). So the whole per-factor loss is charged to the term 0.0837δ of H_{head} . After this charge the remaining product is $\prod_{i \leq 2k, N \nmid i} |1 - \zeta^i|$, which splits into M' full periods, each equal to $\prod_{i=1}^{N-1} |1 - \zeta^i| = N \geq 1$, and one partial period $\geq P_N$; so the full periods contribute $N^{M'} \geq 1$ and the 0.959 is not charged a second time. For $i = Nm$: $|1 - e^{-Nmt}| \geq 0.99Nmr$. With $M' = \lfloor 2k/N \rfloor \leq$

$2\delta/(Nr)$ and $\log M'! \geq M' \log M' - M'$, the function $M'(1 + \log \frac{1}{0.99NrM'})$ is increasing on the range, which gives the second term. Numerically at $N = 16$: $0.01148 + 0.00514$; both terms decrease in N . $\square$

Lemma 6 (landscape; PROVED for $N \geq 16$). *Let $N \geq 16$, $\theta = \theta^*$, $W_\mu = \text{Re}(\Omega_\mu e^{-i\theta})$, and x_A any point of the ray $\arg x = \theta$ with $|x_A| \in [\delta/2, 2\delta]$. Then:*

- (a) *For $\mu \in \mathcal{M} \setminus \{n, n+s\}$ with $|\mu| \leq 4N+1$: $W_\mu \leq T - \eta_N$ on Π_μ , $\eta_N = \min(16.2/N^2, 0.1)$.*
- (b) *For $\mu \in \{n, n+s\}$: on $[x_A, x_\mu]$, $W_\mu(x) - T \leq -0.12|x - x_\mu|^2/\Delta^2$ for $|x - x_\mu| \leq \Delta/2$ and ≤ -0.024 otherwise ($\Delta = |x_\mu - x_A| \leq 0.88$); on the ray, $W_\mu - T \leq -0.99v^2$.*
- (c) *On $\gamma_\mp$ (from x_A), for the remainder modes $\mu_\mp = \nu_\mp N/N'$: $W_{\mu_\mp} \leq 2\delta\ell + \frac{\pi^2}{3N^2} \leq 0.035$; the kernels are $\leq (1 - e^{-(\pi/2)\tan(\pi/8)})^{-1} \leq 2.09$.*
- (d) *Every point of these paths has $\text{Re } x \geq 0.45\delta$.*

The same holds with θ in a neighbourhood $|\theta - \theta^| \leq \varepsilon_N$, with T replaced by $T(\theta) = \max(h_n(\theta), h_{n+s}(\theta))$ and the margins halved.*

Proof. Write $W_\mu(x) = h_\mu - \text{Re}((x - x_\mu)^2 e^{-i\theta}) - \text{Re}(R_\mu e^{-i\theta})$ (Lemma 1(iii)). On the segment put $x = x_\mu - \sigma(x_\mu - x_A)$, $0 \leq \sigma \leq 1$, and $D = \text{Re}((x_\mu - x_A)^2 e^{-i\theta})$. Since $x_A e^{-i\theta} = |x_A| > 0$, $D = \text{Re}(L_\mu^2 e^{-i\theta})/4 - |x_A|\ell + O(|L_\mu||\xi| + |x_A|^2)$, and $\text{Re } x$ increases along the segment from $\text{Re } x_A \geq \frac{\delta}{2} \cos \theta$ to $\text{Re } x_\mu \geq \ell/2 - 10^{-7}$. Always $|\frac{\pi^2}{6} - \Lambda(2Nx)| \leq \frac{\pi^2}{3}$, so $W_\mu(x_A) \leq |x_A|\ell + \frac{\pi^2}{3N^2} \leq 0.0345$.

(a) If $D \geq 0$: $W_\mu \leq h_\mu + |R_\mu| \leq T - \frac{2\pi^2 s^2 \cos \theta^*}{N^2} + \frac{1.647}{N^2} \leq T - \frac{16.2}{N^2}$, using the crude bound of Lemma 1(iii) with $|x - x_\mu| \leq |L_\mu|/2 + |\xi| + 2\delta \leq \frac{1}{2}\sqrt{1.39^2 + (9\pi + \frac{2\pi}{16})^2} + 0.02 \leq 14.4 \leq 15$ (as $|\mu| \leq 4N+1$) and $1.1|w|(1+30N) \leq 1.3 \cdot 10^{-3}$. If $D < 0$: $W_\mu \leq h_\mu - D + |R_\mu| = W_\mu(x_A) + \text{Re}(R_\mu(x_A)e^{-i\theta}) + |R_\mu(x)| \leq 0.0345 + \frac{3.3}{N^2} \leq 0.048 \leq T - 0.1$. On the ray, $W_\mu - h_\mu = -v^2 + O(v^2 e^{-0.85N})$.

(b) Here $h_\mu = T$, $D \geq T - \frac{\pi^2}{6N^2} - 2\delta\ell - 10^{-6} \geq 0.1217$, and $|L_\mu| \leq \sqrt{\ell^2 + (0.4594\ell + 2\pi/16)^2} \leq 1.727$, so $\Delta \leq 0.88$. For $\sigma \leq \frac{1}{2}$ the Taylor bound with $\text{Re } y \geq 0.2137$ gives $|R_\mu| \leq 2\sigma^2 \Delta^2 / (e^{6.84} - 1) \leq 0.0017\sigma^2$, hence $W_\mu - T \leq -\sigma^2(D - 0.0017) \leq -0.12\sigma^2$. For $\sigma \geq \frac{1}{2}$ the crude bound gives $W_\mu - T \leq -D/4 + 1.647/N^2 \leq -0.024$.

(c) With $x = x_A + \varrho e^{i(\theta - \pi/8)}$: $\text{Re}(xL_\mu e^{-i\theta}) = |x_A|\ell + \varrho(\ell \cos \frac{\pi}{8} + \psi_\mu \sin \frac{\pi}{8})$, and $\text{Re}(x^2 e^{-i\theta}) = |x_A|^2 \cos \theta + 2|x_A|\varrho \cos(\theta - \frac{\pi}{8}) + \varrho^2 \cos(\theta - \frac{\pi}{4}) \geq 0.34\varrho^2$ for $|\theta| \leq 24.7^\circ$. The remainder mode has $\psi_{\mu_-} \leq \varphi - 4\pi \leq -3\pi$, so the linear coefficient is $\leq 1.281 - 3.608 < 0$. Symmetrically on γ_+ with $\psi_{\mu_+} \geq 3\pi$. The kernel bound is (iv) of **lem:qi-bounds** with $\alpha_{\min} = \pi/8$.

(d) Clear from the geometry. The perturbation in θ changes every quantity continuously, and all inequalities above are strict. $\square$

VERIFIED independently (sampled, floating; **P1f_landscape_check.py**): for every a/N of the family with $N \in \{16, 17, 20, 23\}$ the sampled maxima satisfy (a)–(c) with room (e.g. 7/17: nondominant $W - T \leq -0.068$, dominant $(W - T)/\sigma^2 \leq -0.43$, remainder ≤ -0.43). The same sampled check also passes for every family member with $5 \leq N \leq 13$, where Lemma 6 is *not* proved (its crude constants need $N \geq 16$).

Lemma 7 (landscape at the shifted heights, N even; PROVED for $N \geq 16$). *Let $N \geq 16$ be even, $\theta = \theta^*$ (so $s = 2$), $\mathcal{M}' = \mathcal{M} + 1$ (the modes μ with $G^*(\mu \mp a) \neq 0$; a is odd), and $T' := h_{n+1}(\theta^*)$. Then*

$$T' = \frac{\ell^2}{4 \cos \theta^*} + \text{Re}(E_w e^{-i\theta^*}) = T + \frac{\pi^2 \cos \theta^*}{N^2}, \quad h_\mu(\theta^*) \leq T' - \frac{4\pi^2 \cos \theta^*}{N^2} \quad (\mu \in \mathcal{M}' \setminus \{n+1\}),$$

and, with x_A as in Lemma 6:

(a') For $\mu \in \mathcal{M}' \setminus \{n+1\}$ with $|\mu| \leq 4N+1$: $W_\mu \leq T' - \eta'_N$ on Π_μ , $\eta'_N = \min(34.2/N^2, 0.1)$.

(b') For $\mu = n+1$: $\Delta = |x_\mu - x_A| \leq 0.78$; on $[x_A, x_\mu]$, $W_\mu(x) - T' \leq -0.159|x - x_\mu|^2/\Delta^2$ for $|x - x_\mu| \leq \Delta/2$ and ≤ -0.033 otherwise; on the ray, $W_\mu - T' \leq -0.99v^2$.

(c') On $\gamma_\mp$, for the remainder modes $\mu_\mp$ of $\mathcal{M}'$ (Section 4 with ϵ replaced by $-\epsilon$): $W_{\mu_\mp} \leq 0.035 \leq T' - 0.11$, kernels ≤ 2.09 .

(d') As Lemma 6(d). The same holds for $|\theta - \theta^*| \leq \varepsilon_N$ with $T'(\theta) = h_{n+1}(\theta)$ and the margins halved.

Proof. Heights: by the choice of θ^* , $\psi_{n+1} - \ell \tan \theta^* = \psi_n + \frac{2\pi}{N} - (\psi_n + \frac{2\pi}{N}) = 0$, and for $\mu = n+1+2k$ it is $\frac{4\pi k}{N}$; Lemma 1(ii) gives $h_{n+1+2k} = T' - \frac{4\pi^2 k^2 \cos \theta^*}{N^2}$, while $T = h_n = T' - \frac{\pi^2 \cos \theta^*}{N^2}$ ($k = -\frac{1}{2}$). The numerical facts of Section 2 are the $s = 2$ worst case: $\cos \theta^* \geq 0.9087$, $|w| \leq 2.3 \cdot 10^{-6}$, and $T' \geq T \geq 0.1499$.

(a') Exactly as Lemma 6(a), with T' in place of T . If $D \geq 0$: $W_\mu \leq h_\mu + |R_\mu| \leq T' - \frac{4\pi^2 \cdot 0.9087}{N^2} + \frac{1.647}{N^2} = T' - \frac{35.874 - 1.647}{N^2} \leq T' - \frac{34.2}{N^2}$ (same crude bound, $|x - x_\mu| \leq 14.4$ for $|\mu| \leq 4N+1$). If $D < 0$: $W_\mu \leq W_\mu(x_A) + |R_\mu(x_A)| + |R_\mu(x)| \leq 0.0345 + \frac{3.3}{N^2} \leq 0.0475 \leq T' - 0.1024$; the bound for $W_\mu(x_A)$ is mode-independent because $x_A e^{-i\theta} = |x_A|$ and $\text{Re } L_\mu = \ell$. On the ray, $W_\mu - h_\mu = -v^2 + O(v^2 e^{-0.85N})$. So $\eta'_N = \min(34.2/N^2, 0.1)$; at $N = 16$, $34.2/256 = 0.134 > 0.1$.

(b') Now $h_\mu = T'$ and $\text{Re}(L_{n+1}^2 e^{-i\theta^*})/4 = \frac{\ell^2}{4 \cos \theta^*} \geq \frac{\ell_0^2}{4} = 0.18268$. Expanding $(x_\mu - x_A)^2 e^{-i\theta}$ as in Lemma 6, with $\text{Re}(x_A^2 e^{-i\theta}) = |x_A|^2 \cos \theta \geq 0$ and $2 \text{Re}(x_\mu x_A e^{-i\theta}) = |x_A|(\ell - 2 \text{Re } \xi)$, gives $D \geq 0.18268 - 2\delta\ell - 10^{-6} \geq 0.18268 - \frac{1.38630}{64} - 10^{-6} \geq 0.1610$ ($\ell \leq \log 4$, $2\delta \leq \frac{1}{64}$). Also $|L_{n+1}| = \ell / \cos \theta^* \leq 1.3863/0.9087 \leq 1.5256$, so $\Delta \leq 0.7628 + |\xi| + 2\delta \leq 0.78$. For $\sigma \leq \frac{1}{2}$: every $y \in [x, x_\mu]$ has $\text{Re } y \geq \frac{1}{2} \text{Re } x_\mu \geq \frac{\ell_0}{4} - 10^{-7} \geq 0.2137$, so $|R_\mu| \leq 2\sigma^2 \Delta^2 / (e^{32 \cdot 0.2137} - 1) \leq 2(0.78)^2 \sigma^2 / 933 \leq 0.0014 \sigma^2$ and $W_\mu - T' \leq -\sigma^2(0.1610 - 0.0014) \leq -0.159\sigma^2$. For $\sigma \geq \frac{1}{2}$: $W_\mu - T' \leq -\frac{D}{4} + \frac{1.647}{N^2} \leq -0.04025 + 0.00644 \leq -0.033$. On the ray $|R_\mu| \leq 2v^2 / (e^{16\ell_0} - 1) \leq 4 \cdot 10^{-6} v^2$.

(c') Lemma 6(c) does not use the height: $W_{\mu_\mp} \leq 2\delta\ell + \frac{\pi^2}{3N^2} \leq 0.035$ with linear coefficient $\ell \cos \frac{\pi}{8} + \psi_{\mu_\mp} \sin \frac{\pi}{8} < 0$ because $|\psi_{\mu_\mp}| \geq 8\pi - \pi$ for the remainder modes $|\mu_\mp| \geq 4N$. And $0.035 \leq 0.1499 - 0.11 \leq T' - 0.11$. $\square$

VERIFIED (P1f_landscape_check.py, column S, run 2026-09-24): for all 250 family members with $16 \leq N \leq 40$ and all explicit modes $|\mu| \leq 4N+1$ (N even) the sampled maxima are $\max(W - T')N^2 \leq -37.69$ over non-dominant modes (proved: ≤ -34.2), dominant $(W - T')/\sigma^2 \leq -0.187$, remainder ≤ -0.175 ; no violation. The column B (Lemma 6, now with $|\mu| \leq 4N+1$ for N even) gives $\max(W - T)N^2 \leq -19.65$ (proved: ≤ -16.2), dominant ≤ -0.174 .

6 The two-saddle asymptotic and the zeros

Theorem 8 (PROVED). Let $N \geq 16$ and ζ in the family. Put $\kappa = C_\zeta \sqrt{(1-w)/(2N(1+w))}$, $C_\zeta = \prod_{i < N} (1 - \zeta^i)^{-(\frac{1}{2} - \frac{i}{N})}$. There are $\varepsilon_N, \eta'_N > 0$ such that, uniformly for $|\theta - \theta^*| \leq \varepsilon_N$ as $r \rightarrow 0$,

$$B(\zeta e^{-t}) = \kappa(A(n)e^{V_n/t}(1 + O(t)) + A(n+s)e^{V_{n+s}/t}(1 + O(t))) + O(e^{(T(\theta) - \eta'_N)/r}).$$

Proof. Section 4 writes B exactly as H plus explicit-mode integrals over Π_μ^ρ plus kernel remainders over $\gamma_\mp^\rho$. By Lemma 4 (with $\delta' = 0.45\delta \cos \theta$, Lemma 6(d)), $|g_\rho e^{2\pi i \nu s}| \leq C_N r^{1/2} e^{W_\mu(x)/r + O(r(1+|x|))}$ on all paths, and $ds = dx/(N't)$. Lemma 5 bounds $|H| \leq (K+1)e^{(0.0167+o(1))/r}$, Lemma 6(a),(c) bound all non-dominant explicit modes and both remainders by $C_N r^{-2} e^{(T - \eta_N)/r}$ (the Gaussian decay on the infinite pieces and the kernel bound 2.09). For $\mu \in \{n, n+s\}$, Lemma 6(b) localises the integral at x_μ ; the path enters from the valley of x_A (since $D > 0$, the direction $x_A - x_\mu$ lies in the descent sector)

and leaves along $e^{i\theta/2}$, so Laplace's method with a complex large parameter, uniform for $\arg t$ in a compact subset of $(-\frac{\pi}{2}, \frac{\pi}{2})$ [1], gives

$$\frac{1}{N't} \int_{\Pi_\mu^\rho} F_\rho(x/t) e^{2\pi i \mu x/(Nt)} e(-\frac{\mu\rho}{N}) dx = \frac{1}{N'} \sqrt{\frac{N(1-w)}{2(1+w)}} C_\zeta e(-\frac{\mu\rho}{N}) e^{a_\rho(x_\mu) - \log C_\zeta} e^{V_\mu/t} (1 + O(t)),$$

with $\Omega_\mu'' = -2(1+w)/(1-w)$ and the principal root. Summing over $\rho < N'$ (for N even, $\sum_{\rho < N'} = \frac{1}{2} \sum_{r < N}$ on $\mathcal{M}$, because the r and $r + N'$ summands agree when $\mu \in \mathcal{M}$) gives $\kappa A(\mu) e^{V_\mu/t} (1 + O(t))$ with $A(\mu)$ as in Lemma 2. Continuity in θ (last sentence of Lemma 6) gives the uniformity. $\square$

Theorem 9 (zeros of B at ζ ; PROVED). *Let $N \geq 16$ and $\zeta = e(a/N)$, $\frac{1}{5} \leq \frac{a}{N} \leq \frac{4}{5}$. Put $\Delta V = V_{n+s} - V_n = \frac{\pi i s}{N} (L_n + \frac{i\pi s}{N})$, $\varrho = A(n+s)/A(n) = \frac{G^*(n+s)}{G^*(n)} e^{i\pi s \ln/N}$, and let u_j be the solutions of $1 + \varrho e^{\Delta V u} = 0$ with $\operatorname{Re} u > 0$, $\arg u \rightarrow -\theta^*$, ordered by modulus, $t_j^0 = 1/u_j$. There is j_0 such that for $j \geq j_0$, $B(\zeta e^{-t})$ has exactly one zero t_j^* in $D_j = \{|1/t - u_j| < 1/|\Delta V|\}$; it is simple and $t_j^* = t_j^0 + O(j^{-3})$. So $q_j^* = \zeta e^{-t_j^*} \rightarrow \zeta$ along $\arg t = \theta^*$, the conjugates tend to $\bar{\zeta}$, and every q_j^* , $\bar{q}_j^*$ is a simple pole of $G_0^T = \Sigma_0/(1 - \Sigma_1)$.*

Proof. $A(n) \neq 0$ (Lemma 2, Theorem M). On D_j , $\arg t = \theta^* + O(1/j)$ and $|\operatorname{Re}(\Delta V/t)| \leq 1$, so Theorem 8 applies and $B/(\kappa A(n) e^{V_n/t}) = g_0(u) + O(1/j)$, $g_0(u) = 1 + \varrho e^{\Delta V u} = 1 - e^{\Delta V(u - u_j)}$, using $|\varrho e^{\Delta V u}| \leq e$ on D_j and $e^{(T(\theta) - \eta'_N)/r} / |e^{V_n/t}| \leq e^{1 - \eta'_N/r}$. $\operatorname{Re}(\Delta V e^{-i\theta^*}) = 0$ by the choice of θ^* , so the zeros u_j lie on a line with $\arg u \rightarrow -\theta^*$, spaced $2\pi/|\Delta V|$ apart, more than the diameter $2/|\Delta V|$ of D_j ; $|g_0| > 0.28$ on ∂D_j . Rouché as in Step 5 of `prop:minusone`. Simple poles of G_0^T : Step 6 of `prop:minusone` ($\Sigma_0 S_0 = 2q/(1 - q) \neq 0$ at every zero of $1 - \Sigma_1$ in $0 < |q| < 1$), and real coefficients for the conjugates. $\square$

VERIFIED (`P1f_zeros.py`, floating, 30–300 digits): $|1/t_j^* - u_j| |\Delta V| = 0.0018, 0.00036, 0.00014$ at $j = 8$ for $3/10, 4/11, 7/17$, decreasing like $|t_j|$; the tie rays are $\theta^* = 0^\circ, 6.31^\circ, 3.92^\circ$.

Corollary 10 (density of the poles of G_0^T ; PROVED). *The set of accumulation points on $|q| = 1$ of the poles of G_0^T contains the closed arc $\{\arg q \in [\frac{2\pi}{5}, \frac{8\pi}{5}]\}$. Equivalently, the poles of $G_0^T(x^2)$ accumulate at every point of $\arg x \in [\frac{\pi}{5}, \frac{4\pi}{5}] \cup [\frac{6\pi}{5}, \frac{9\pi}{5}]$.*

Proof. The accumulation set is closed. By Theorem 9 it contains every $e(a/N)$ with $N \geq 16$, $\frac{1}{5} \leq \frac{a}{N} \leq \frac{4}{5}$, and these are dense in the arc. $\square$

7 Transfer to U and V

$S_\pm = \sum_k b_k q^{\pm k}$, $S_e = S_+$, $t_1 = -\frac{1}{2}(1 + S_-/S_e)$ on $|q| < 1$ wherever $S_e \neq 0$ (Step 0 of `prop:t1minusone`).

Theorem 11 (PROVED). *Let $N \geq 16$, ζ in the family, and q_j^* as in Theorem 9. (i) $S_\pm(q_j^*) = 2\kappa A(n) \frac{Z_N^\pm}{Z_N} e^{\mp x_n} \frac{G^*(n \mp a)}{G^*(n)} e^{V_n/t_j^*} (1 + O(1/j))$ for N odd, and $S_\pm(q_j^*) = \kappa A^\pm(n+1) e^{V_{n+1}/t_j^*} (1 + O(1/j))$ for N even. In particular $S_e(q_j^*) \neq 0$ for large j if and only if $Z_N^+ \neq 0$. (ii) If $Z_N^+ \neq 0$: $t_1(q_j^*) = t_1^* + O(1/j)$, $t_1^* = -\frac{1}{2}(1 + \omega)$, $\omega = \frac{Z_N^-}{u_0 Z_N^+}$. (iii) If moreover $\omega \notin \left\{ -1 - \frac{2(1-\zeta^2)}{\zeta}, -1 - \frac{2(1-\bar{\zeta})}{\zeta} \right\} \cup \left\{ \frac{1}{x}, \frac{1+\zeta}{x(1-x+\zeta)} - 1 : x = \pm\sqrt{\zeta} \right\}$, then $1 - \mathbf{g}_U t_1, 1 - \mathbf{g}_V t_1, \Pi_1, \Pi_q$ have non-zero limits at q_j^* for both roots, and for large j , $\pm\sqrt{q_j^*}, \pm\sqrt{\bar{q}_j^*}$ are poles of U and of V ; these tend to $\pm\sqrt{\zeta}, \pm\sqrt{\bar{\zeta}}$. (Standing: Theorems `thm:model`, `thm:RJ`, as in `cor:Uminusone`.)*

Proof. (i) The k -th term of $S_{\pm}$ is $b_k \zeta^{\pm k} e^{\mp tk}$; in the class decomposition $\zeta^{\pm(N'j+\rho)} = \zeta^{\pm\rho}(\pm 1)^{\dots}$, which multiplies F_{ρ} by $\zeta^{\pm\rho} e^{\mp x}$ and, for N even, flips ϵ (as $\zeta^{N'} = -1$). So the proof of Theorem 8 applies verbatim with amplitude $A^{\pm}(\mu)$ of Lemma 2 and modes μ with $G^*(\mu \mp a) \neq 0$: for N odd the same pair $n, n+1$ is dominant on θ^* ; for N even the single mode $n+1$, of height $T' = T + \pi^2 \cos \theta^*/N^2$, with the others lower by at least $4\pi^2 \cos \theta^*/N^2$; the landscape bounds at these heights are Lemma 7 (used in place of Lemma 6; $|e^{\mp x}| \leq e^{|x|}$ is bounded on the finite pieces and dominated by the Gaussian on the rays). The error term is $O(e^{(T'-\eta_N/2)/r})$, relatively $O(e^{-\eta_N/(2r)})$. For N odd insert $e^{\Delta V/t_j^*} = -A(n)/A(n+1) + O(1/j)$ (Theorem 9): $S_{\pm} \propto A^{\pm}(n) - A^{\pm}(n+1)A(n)/A(n+1) = A^{\pm}(n)[1 - e^{\mp i\pi/N} R_{\pm}^{-1}]$, with $R_{\pm} = \frac{G^*(n \mp a)G^*(n+1)}{G^*(n+1 \mp a)G^*(n)}$. By Lemma 1(iv), $R_{\pm} = e(\mp 4a \cdot 2a/N) = e(\mp 2/N) = -e(\mp \frac{1}{2N})$, since $\bar{2} = \frac{N+1}{2}$. So $e^{\mp i\pi/N} R_{\pm}^{-1} = -1$ and the bracket is exactly 2. (ii) $S_-/S_e = \frac{A_-}{A_+}$ at the dominant mode μ ($\mu = n$ or $n+1$) $= e^{2x_{\mu}} \frac{G^*(\mu+a)}{G^*(\mu-a)} \frac{Z_N^-}{Z_N^+}$. *Claim:* $G^*(\mu+a) = e(-\mu/N) G^*(\mu-a)$ for every N and every $\mu \in \mathbb{Z}$. Substitute $r \mapsto r+1$ (a bijection of $\mathbb{Z}/N$) in $G^*(\mu+a) = \sum_r e((ar^2 - (\mu+a)r)/N)$: the exponent becomes $ar^2 + 2ar + a - (\mu+a)r - (\mu+a) = ar^2 - (\mu-a)r - \mu$, so $G^*(\mu+a) = e(-\mu/N) \sum_r e((ar^2 - (\mu-a)r)/N) = e(-\mu/N) G^*(\mu-a)$. This covers N odd, $N \equiv 2 \pmod{4}$ and $N \equiv 0 \pmod{4}$ at once; no CRT factorisation is needed (the earlier text appealed to one without writing it). At the dominant mode, $\mu \mp a \in \mathcal{M}$ ($\mu = n$ or $n+1$, N odd; $\mu = n+1 \in \mathcal{M}'$, N even), so $G^*(\mu-a) \neq 0$ by Lemma 1(iv) and $\frac{G^*(\mu+a)}{G^*(\mu-a)} = e(-\mu/N)$. (Numerically re-checked for all a/N in the family, $N \leq 40$, all μ with $G^*(\mu-a) \neq 0$: VERIFIED.) And $e^{2x_{\mu}} = c e(\mu/N)(1-w)^{-2/N} = e(\mu/N)/u_0$. Hence $S_-/S_e \rightarrow \omega$. (iii) As in `cor:Uminusone/cor:Ui`: the first two paragraphs of the proof of `cor:Uminusone` need only $0 < |q_j^*| < 1$, $S_e(q_j^*) \neq 0$ and $1 - \mathbf{g}t_1 \neq 0$; $\mathbf{g}_U \rightarrow \zeta/(1-\zeta^2)$, $\mathbf{g}_V \rightarrow \zeta/(1-\zeta)$ are finite as $\zeta \neq \pm 1$; the listed values of ω are exactly the zeros of $1 - \mathbf{g}_U t_1^*$, $1 - \mathbf{g}_V t_1^*$, $t_1^* + \frac{1+x}{2x}$ and $t_1^* \frac{1-x+\zeta}{1+\zeta} + \frac{1}{2x}$ in the closed forms `eq:RJclosed1`, `eq:RJclosedq` of paper2a (Theorem `thm:RJ`), whose remaining factors are finite and non-zero for $x = \pm\sqrt{\zeta} \neq \pm 1$, $\zeta \neq -1$. $\square$

VERIFIED: `P1f.t1.py` finds $|t_1(q_j^*) - t_1^*|/|t_j^*| \approx 0.91, 1.44, 1.53$ (stable in j) at $3/10, 4/11, 7/17$. At $1/4$ the formula gives $\omega = -1.1454972 + 1.1454972i$ and $t_1^* = 0.072748612 - 0.57274861i = \frac{(\sqrt{15}-3) - (\sqrt{15}+3)i}{12}$, and $\Pi_1 \rightarrow -3.0048 + 2.9237i$, $\Pi_q \rightarrow -2.7091 + 3.3891i$ at $x \rightarrow e^{i\pi/4}$: the values of `cor:Ui`, obtained there by a different route.

Proposition 12 (computer-assisted). *For every reduced a/N with $3 \leq N \leq 150$, $N \leq 5a \leq 4N$, $a/N \neq \frac{1}{2}$ (4116 cases), $Z_N \neq 0$, $Z_N^+ \neq 0$ and the six ω -conditions of Theorem 11(iii) hold; the certified moduli of Z_N , Z_N^+ and of the five transfer factors (both roots) are ≥ 0.2331 .*

Script `P1f.transfer-cert.py`: $Z, Z^{\pm}$ by the exact finite formula $\sum_k \gamma_k^{(\pm)} e^{\Phi(\zeta^k)}$ with $\gamma_k^{(\pm)} = \frac{1}{N} \sum_{\nu} \zeta^{-\nu^2 \mp \nu - k\nu}$ and a rigorous tail; fails loudly; blocks 3–70, 71–105, 106–130, 131–150 all CERT PASSED.

Corollary 13 (computer-assisted). *For every $\zeta = e(a/N)$ with $16 \leq N \leq 150$ and $\frac{1}{5} \leq \frac{a}{N} \leq \frac{4}{5}$, the poles of U and of V accumulate at $\pm\sqrt{\zeta}$ and $\pm\sqrt{\bar{\zeta}}$ (standing: Theorems `thm:model`, `thm:RJ`).*

8 What is not proved

1. **Density of the poles of U and V on the arcs (OPEN).** Theorem 11 reduces it to: for a dense set of a/N in $[\frac{1}{5}, \frac{4}{5}]$, $Z_N^+(a/N) \neq 0$ and $\omega(a/N)$ avoids six explicit values. $Z_N^{\pm} = Z_N(Y)$ at $Y = \zeta^{\mp 1}$ is the generating function of Theorem 2 of P1d at a rotated point; Theorem M does not cover it. The Lemma A regime of P1e ($|Z^{\pm} - 1| \leq e^{\mathcal{A}} - 1$) covers only a nowhere dense set of a/N , since every rational j/m has a neighbourhood of radius $\asymp |u_0|^m/m^2$ where $\mathcal{A} \geq \log 2$; a hazard induction for $Z^{\pm}$ and a quantitative two-sided control of ω inside hazards would be

needed. The certificate (4116 cases, margin ≥ 0.233 , the margin essentially constant in N) is strong evidence (VERIFIED) but not a proof for $N > 150$.

2. **Theorem 9** for $3 \leq N \leq 15$ is not proved here (except $N = 2$, $N = 4$: `prop:minusone`, `prop:qi`). The crude perturbation constant $\frac{\pi^2}{6}N^{-2}$ in Lemma 6(b) needs $N \geq 10$ (with thin margins) and is taken at $N \geq 16$; a sampled check passes for all $5 \leq N \leq 13$ (VERIFIED). Not needed for Corollary 10.
3. The $O(t)$ of Laplace's method and j_0 are not explicit (as in paper2a).

Files (all in `rootunity_zeros/general/P1f/` unless noted). `P1f_model.py` (saddle model vs. B), `P1f_zeros.py` (zeros vs. prediction), `P1f_t1.py` (t_1 , transfer factors at the zeros), `P1f_zpm.py` ($Z^\pm$, floating), `P1f_transfer_cert.py` (Proposition 12, Arb, fail-loud), `P1f_landscape_check.py` (sampled check of Lemmas 6 and 7; output for $16 \leq N \leq 40$ in `ls_even4N.txt`), `P1f_certlog.txt` (raw outputs); `../dround.py`, `../lowseed_directed.txt`, `../base_arc_*.txt` (Task A). Every run used `perl -e 'alarm 290; exec @ARGV'` and stayed below 300 MB.

References

- [1] F. W. J. Olver, *Asymptotics and Special Functions*, Academic Press, 1974.